

Octave renormalization

intra-block $q^\dagger$ is not a Cauchy sequence

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Abstract

The named selected family has constant Δ on each $r = \lfloor \sqrt{k} \rfloor$ block. The live arithmetic source is $n < a_k$ (tents vanish for $n \geq a_k$, r465), not the a^2 -octave $(a_k^2, a_{k+1}^2]$. The active-octave weight W and its unconditional Chebyshev bound both *grow*. Isolating the new primes at fixed geometry can move $q^\dagger$ by 0.22 while the total step is -0.012 . Intra-block Δq oscillates through $k = 15$. Two Δ -halving edges are positive $(+0.045, +0.018)$ — not a law. Without a proved summable rate the proposed cut “ $q(\text{block-limit } r) < 1$ for infinitely many r ” is an alias of the original quantifier. No RH claim.

Status. No new SATZ. Ausgang OCTAVE_NOT_CAUCHY / EDGE_TWO_POINTS / REDUCTION_ALIAS. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.

1 Geometry

Lemma 1 (r -sequence). $r_k = \lfloor \sqrt{k} \rfloor$, $\Delta_k = 2^{-r_k} \log 2$, $m_k = k 2^{r_k} - 1$. In the measured range: $k = 5..8$ have $r = 2$; $k = 9..15$ have $r = 3$; $k = 16$ has $r = 4$. Block edges sit at $k = (r + 1)^2$ ($8 \rightarrow 9$, $15 \rightarrow 16$). The step $9 \rightarrow 10$ is intra-block ($r = 3 \rightarrow 3$), not an r -jump.

Lemma 2 (active source). On this mesh $M\Delta = \log a$, so every tent with $\log n \geq \log a$ vanishes on lags $0, \dots, M - 1$ (r465). The live increment $k \rightarrow k + 1$ is the octave $n \in [2^k, 2^{k+1})$, plus 2^{r-1} new degrees and a longer Archimedean grid. The a^2 -tail is a completeness label, not a perturbation of c^P .

2 Octave perturbation

Unconditional Rosser–Schoenfeld: $\psi(x) < 1.03883x$. Then

$$\sum_{2^k \leq n < 2^{k+1}} \frac{\Lambda(n)}{\sqrt{n}} \leq \frac{\psi_{\text{up}}(2^{k+1}) - \psi_{\text{lo}}(2^k)}{2^{k/2}},$$

which *grows* in k .

$k \rightarrow k+1$	Δq	Δq_{oct}	W	bound	dN	type
5 $\rightarrow$ 6	+0.00746	+0.0669	4.31	11.8	2	intra
6 $\rightarrow$ 7	−0.02486	+0.1437	6.70	10.5	2	intra
7 $\rightarrow$ 8	−0.00833	+0.0417	9.15	14.5	2	intra
8 $\rightarrow$ 9	+0.04542	+0.0989	13.65	20.1	20	BLOCK
9 $\rightarrow$ 10	+0.03439	+0.0464	18.63	28.0	4	intra
10 $\rightarrow$ 11	−0.01180	+0.2199	26.31	39.1	4	intra
11 $\rightarrow$ 12	−0.00624	+0.0994	37.81	54.7	4	intra
12 $\rightarrow$ 13	+0.01185	+0.0165	52.68	76.7	4	intra
13 $\rightarrow$ 14	+0.01784	+0.0323	75.34	107.6	4	intra
14 $\rightarrow$ 15	−0.01703	+0.0068	105.7	151.1	4	intra
15 $\rightarrow$ 16	+0.01844	+0.0091	149.9	212.5	68	BLOCK

Δq_{oct} is the new-octave effect at *fixed* $k+1$ geometry. Sharpness bound/ $|\Delta q|$ lies in 400–11 000. Cauchy interlacing on a 2^{r-1} -row pad does not produce a contraction of $q^\dagger = s_N/(s_{N-1} + 5/7)$.

Lemma 3 (not Cauchy). *The octave-weight bound is unconditional and not summable. Measured intra-block Δq oscillates in $[-0.025, +0.034]$ through $k = 15$ and is not dominated by W . `OCTAVE_NOT_CAUCHY`.*

3 Block edges

Lemma 4 (two points). *The only Δ -halvings in range are $8 \rightarrow 9$ (+0.04542) and $15 \rightarrow 16$ (+0.01844). Both signs are +. The ratio 2.46 is not 2. `EDGE_TWO_POINTS`.*

New indices carrying the table were interval-certified by the r465 Chebyshev–Cholesky route ($k = 13, 14, 15, 16$ all have $q_{\text{hi}} < 1$; $k = 16$ took 35 s). $k = 17..20$ were not needed for the verdict (next edge is $k = 25$).

4 The reduced problem

Wording. “ $q^\dagger$ of the r -block limit is < 1 for infinitely many r , using a proved intra-block Cauchy rate.”

What is proved. The r -sequence; vanishing of $n \geq a$ tents; an unconditional growing bound on the active-octave mass.

What is only measured. The Δq table; two edge jumps; certified enclosures at $k = 13..16$.

Alias (r303). A finite block has $2r + 1$ members. Without a rate $\sum |\Delta q| < \infty$ the block limit is not known to exist, and even if it existed one would still need infinitely many living block limits — the same ∞ as infinitely many k , up to a constant factor $2r + 1$. That is a slack shift, not a smaller quantifier. `REDUCTION_ALIAS`.

5 Kill table

Kill	Status
Comb inventory $n < a$ (live) vs $n \leq a^2$ (label)	sealed; $k = 5$ is 17 vs 198
r465 certificates on new verdict k	13, 14, 15, 16 live
ψ -bound vs measurement	bound grows; sharpness 400+
Determinism	two identical $k = 5$ packs
$9 \rightarrow 10$ as r -jump	false; both $r = 3$
Quantifier cut	refused (alias)

Claim boundary. Finite-window census. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.